

ENEL301 – Fundamentals of Engineering Economics and Management

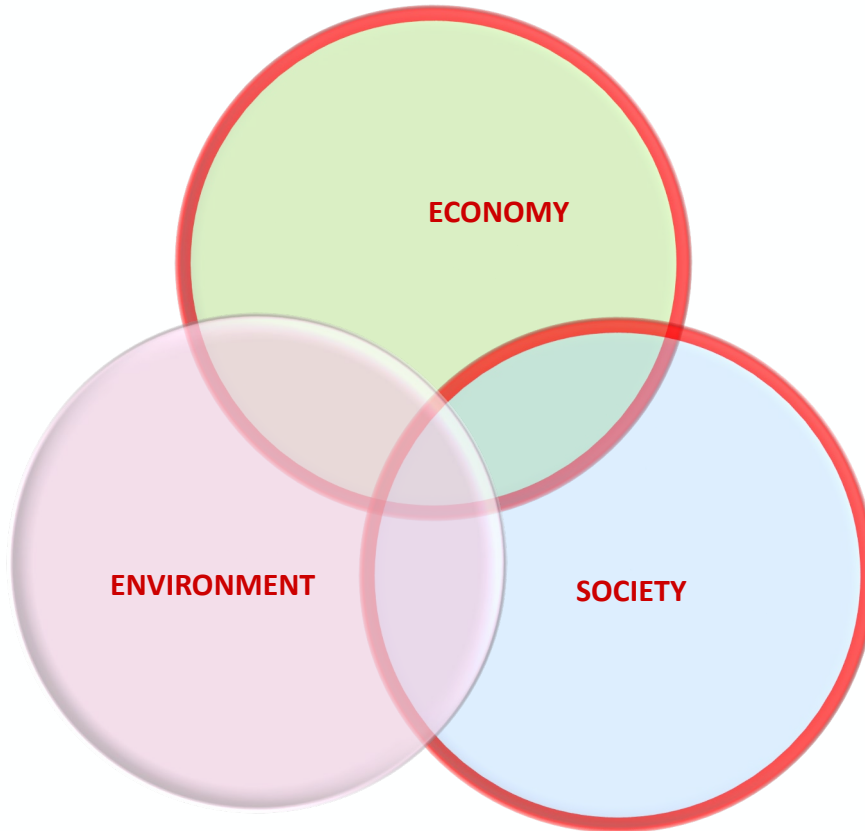
Economic Assessment

Social Assessment

What we are going to cover and when...

Week	Lectures Tuesday 1-2 & Friday 10-11	Tutorial / Workshop Weeks 7-12 Wednesday 3-4 or 4-5
7	Life Cycle Assessment (2/2)	Life Cycle Assessment
	Social and Economic Assessment	
8	Time Value of Money & Cash Flow (1/2)	Economic Assessment, Time Value of Money & Cash Flow
	Time Value of Money & Cash Flow (2/2) Cost Accounting	
9	Financial Accounting & Reporting (1/2)	Cost Accounting, Financial Accounting, Financial Reporting
	Financial Accounting & Reporting (2/2)	
10	Project Management	Project Management
	Risk Management	
11	Opportunity Analysis	Support for Sustainability Assignment
	Integrated Decisions (1/2) Guest Lecturer – Mike Ullrich (Torqos, formerly Fisher & Pykel Technologies)	
12	Integrated Decisions (2/2)	Risk Management and Opportunity Analysis 2
	Course Summary	

What is sustainability?



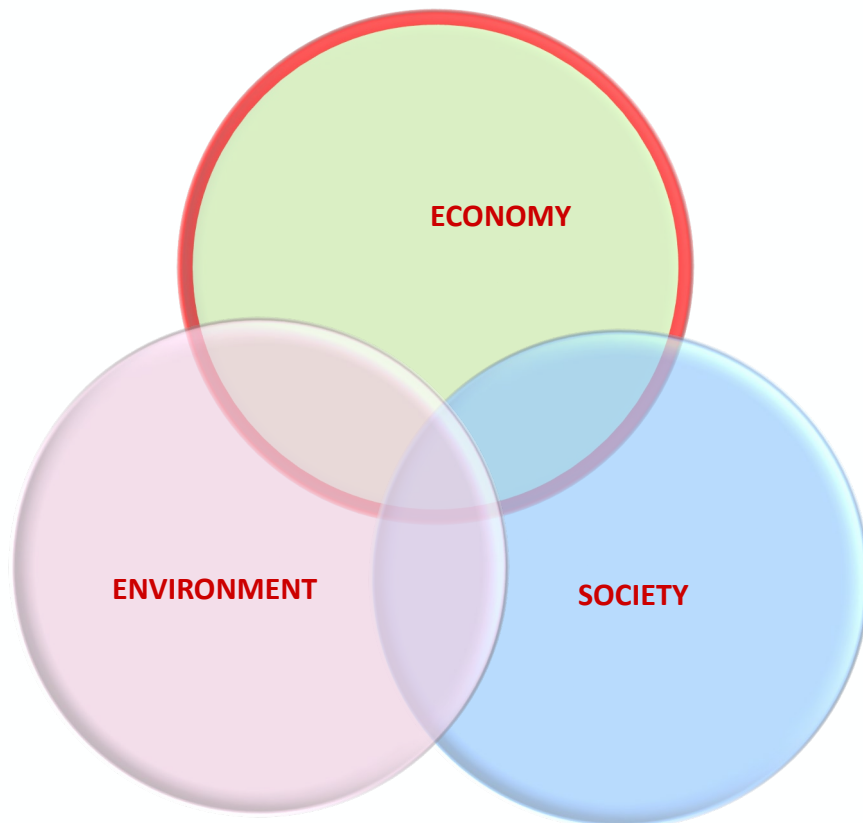
**“..meeting the needs of the present
without compromising the ability of
future generations to meet their own
needs.”**

**Brundtland Report – Our Common
Future**



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Economic assessment



- Local and global markets
- Distribution of goods, services
- Flow of money through society → economy
- Purchasing behaviour
- Supply & demand
- Market growth / decline etc.

NOT Finance

- Assessment of money
- How, on what, when and why should money be spent

(this is coming soon)

What is an economy?

Why is this important?

- Market of buyers and sellers
- Supports determination of economic viability
- Understand connections with other economic sectors
- The flow of money leads to economic, social and environmental outcomes

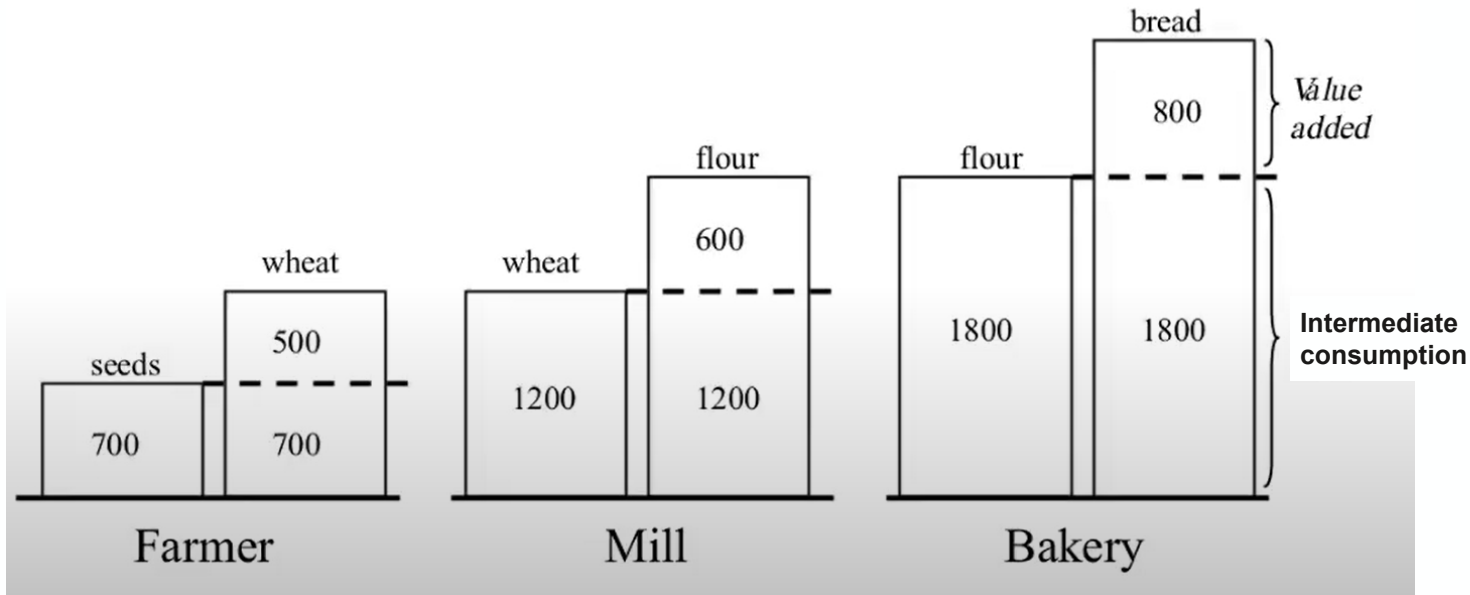


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Cost < Price < Value

Economic principles

GDP (production based)



Economic principles

GDP

- Production-based:
 - Gross value added from production across all sector in the economy
 - To avoid double-counting, must subtract costs of inbound goods/services from value
- Expenditure-based
 - what is spent on the acquisition of products/services.
- Income-based
 - total income earned by households in a year (inc. wages, rent, interest)
- In a closed economy $\text{Production} = \text{Expenditure} = \text{Income}$
(we don't have a closed economy – exports and imports cause differences in these GDP approaches)

Economic principles

GDP – Why ?

- Economic health check!
- Generally (and on average), if inflation is in-check, an increase in GDP indicates that people and businesses are financially doing OK
- In general, there is a correlation between GDP and social outcomes, e.g. life expectancy

What causes GDP to decrease?

- Industry decisions to produce less → households reduce spending
- AND/OR
- Household decide to spend less → less production

Economic principles

Limitations of GDP

- Does not account of loss of capital (e.g. wear and tear)
 - GDP less depreciation = net domestic product
- Does not measure standard of living or well-being (subjective)
- Does not account for environmental damage → value at the expense of environment
- Alternatives to GDP exist to address these, but there is no consensus use, e.g. :
 - **Purchasing Power Parity**
 - **Human Development Index (HDI)**
 - **Gross National Income** (like GDP, but trade is accounted for)
 - **The Big Mac Index** (a variant of PPP)

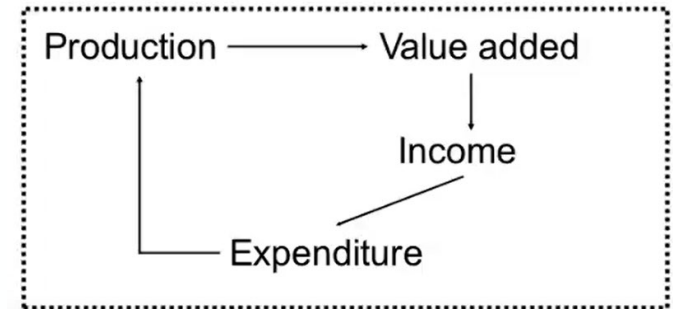


Economic trade

System of National Accounts



When you spend \$, that money will “ripple” through the economy, and some of it will come back to you



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Economic trade

National Accounts Input Output Tables

- From the “System of National Accounts”
- The flows of money between sectors in an economy (e.g. New Zealand)

Useful for understanding:

- effect of policy have on key economic variables
- impact of changes in producer prices or wages
- Imports and exports
- If I spend \$ in a particular sector, how will that money likely flow through the economy?
- <https://www.stats.govt.nz/information-releases/national-accounts-input-output-tables-year-ended-march-2024/>

Economic trade

National Accounts Input Output Tables - Supply

A	B	C	D	E	F	G	H	I	J	K	L
Table 1											
Supply of products in basic prices											
Year ended March 2024											
NZ\$ (millions)											
	Total supply purchasers prices	Margins	Taxes	Total supply basic prices	Horticulture and fruit growing	Sheep, beef cattle, and grain farming	Dairy cattle farming	Poultry, deer, and other livestock farming	Forestry and logging	Fishing and aquaculture	Agriculture, forestry, fishing and services
Sheep	3,000			3,000	35	2,744	87	68			1
Cattle	4,368	11		4,357	85	3,681	387	128	1		62
Other livestock	514	50	9	454		5	1	246	8		
Deer	211			211	3	62	6	97	1		40
Pigs	289	51		238	3	15	9	189			21
Raw milk	16,917			16,917	109	582	14,803	123			51
Wool	460	18		441	5	364	10	8			
Other animal products	1,085	103	57	925	3	36	6	736			93
Non-wood forest products	37			37							
Wood in the rough	5,463	545	4	4,915		8	1		4,557		8
Fish	438		11	427						322	
Crustaceans	665	80	2	583						311	
Support services to agriculture (including animal husbandry)	6,000		1	5,999	89	171	48	38			4,21
Support services to forestry and logging	1,910		1	1,909					710		1,19
Fishing services	365			365						48	242
Coal, coke, and tar products	663		3	660							
Crude petroleum	888		22	866							

- e.g. Forest and Logging industry supplied:
- \$8 million in other livestock, \$4557 million of 'wood in the rough' products, etc.

Economic trade

National Accounts Input Output Tables - Use

	A	L	M	N	O	P	Q	R	S	T	U	
1	Table 2											
2	Use of products in basic prices											
3	Year ended March 2024											
4	NZ\$ (millions)											
5												
6		Metal ore and non-metallic mineral mining and quarrying	Exploration and other mining support services	Meat and meat product manufacturing	Seafood processing	Dairy product manufacturing	Fruit, oil, cereal, and other food product manufacturing	Beverage and tobacco product manufacturing	Textile and leather manufacturing	Clothing, knitted products, and footwear manufacturing	Wood product manufacturing	P and pa me
7	Retail margin	7		113	15	57	234	43	8	27	16	
8	Wholesale margin	38	3	189	55	210	549	205	67	29	184	
9	Living plants, buds, seeds, spice, and aromatic crops						111					
10	Vegetables				18		489	1				
11	Kiwifruit											
12	Pome fruit (apples, pears)						7					
13	Other fruit and nuts						58	892				
14	Forage products, fibres, sugar crops, unmanufactured tobacco						4	35				
15	Cereals			125			198	5				
16	Sheep			2,825								
17	Cattle	1		3,079								
18	Other livestock			44								
19	Deer			164								
20	Pigs			230								
21	Raw milk					16,853	38					

- e.g. Fruit, oil, cereal and other food manufacturing spent:
 - \$111 million on living plants, \$489 million on vegetables etc...

Economic trade

Inter-Country Input-Output (ICIO) tables

Use. By country and industry

		I		U		T	
		Intermediate use (USD Million)		Intermediate use (USD Million)		Intermediate use (USD Million)	
		ARG_A01_02		ARG_A03		ARG_B05_06	
		ARG Argentina A01_02		ARG Argentina A03		ARG Argentina B05_06	
Country	Industry	Agriculture, hunting, forestry		Fishing and aquaculture		Mining and quarrying, energy producing products	
		Intermediate use (USD Million)		Intermediate use (USD Million)		Intermediate use (USD Million)	
		ARG_A01_02		ARG_A03		ARG_B05_06	
		ARG Argentina A01_02		ARG Argentina A03		ARG Argentina B05_06	
Country	Industry	Agriculture, hunting, forestry		Fishing and aquaculture		Mining and quarrying, energy producing products	
ARG	Argentina A01_02	Agriculture, hunting, forestry		Fishing and aquaculture		Mining and quarrying, energy producing products	
ARG	Argentina A03	7093.5057		12.6376		0.8651	
ARG	Argentina B05_06	0		65.5517		0.0186	
ARG	Argentina B07_08	8.3094		0.0529		147.4795	
ARG	Argentina B09	0.363		4.00E-04		32.6632	
ARG	Argentina C10T12	1847.3562		7.3929		2534.46	
ARG	Argentina C13T15	152.7815		4.0246		11.634	
ARG	Argentina C16	47.7602		0.4273		4.8098	
ARG	Argentina C17_18	15.5876		9.3678		0.6532	
ARG	Argentina C19	720.3163		60.0946		3.0655	
ARG	Argentina C20	162.7179		3.1466		2.832	
ARG	Argentina C21	330.1546		2.2216		18.6758	
ARG	Argentina C22	749.5116		3.08		9.2628	
ARG	Argentina C23	30.0788		0.1698		37.3833	
ARG	Argentina C24	56.4227		1.7435		176.3923	
ARG	Argentina C25	35.6461		12.7686		50.3652	
ARG	Argentina C26	10.1435		0.3346		151.0058	
ARG	Argentina C27	46.1537		2.5482		7.3379	
ARG	Argentina C28	104.1418		9.0375		7.6294	
ARG	Argentina C29	147.1208		7.9785		32.9771	
ARG	Argentina C30	3.4201		0.3147		11.9186	
ARG	Argentina C31T33	26.4189		3.9175		2.548	
ARG	Argentina D	67.5709		6.0022		36.6047	
ARG	Argentina E	15.162		3.3709		60.059	
ARG	Argentina F	17.5488		4.00E-04		1.5939	
ARG	Argentina G	5592.0762		100.276		16.2349	
ARG	Argentina H49	3603.3693		3.5803		251.1786	
ARG	Argentina H50	0.1425		0		42.5019	
ARG	Argentina H51	38.173		6.5367		0	
ARG	Argentina H52	29.3672		45.6272		16.206	
ARG	Argentina H53	6.0662		0.6257		50.7519	
ARG	Argentina I	71.5024		9.8923		18.9391	
ARG	Argentina J58T60	14.8106		0.2832		36.8331	
ARG	Argentina J61	106.3638		6.4483		5.6226	

Supply

By country and industry

<https://www.oecd.org/en/data/datasets/inter-country-input-output-tables.html>

(A more useable version is available on LEARN)

Classification: In-Confidence

Economic trade

Inter-Country Input-Output (ICIO) tables

				Intermediate use (\$USD Million)
				NZL_J62_63
				NZL
				New Zealand
				J62_63
				IT and other information services
Country	Country	Industry		
AUS	Australia	H51	Air transport	1.8811
AUS	Australia	H52	Warehousing and support activities for transportation	0.1249
AUS	Australia	H53	Postal and courier activities	0.3639
AUS	Australia	I	Accommodation and food service activities	0.0366
AUS	Australia	J58T60	Publishing, audiovisual and broadcasting activities	1.6063
AUS	Australia	J61	Telecommunications	3.2681
AUS	Australia	J62_63	IT and other information services	14.069
AUS	Australia	K	Financial and insurance activities	0.7171
AUS	Australia	L	Real estate activities	0.1357
AUS	Australia	M	Professional, scientific and technical activities	5.0539
AUS	Australia	N	Administrative and support services	1.1216
AUS	Australia	O	Public administration and defence; compulsory social security	0.5934
AUS	Australia	P	Education	0.1836
AUS	Australia	Q	Human health and social work activities	0
AUS	Australia	R	Arts, entertainment and recreation	0.1002
AUS	Australia	S	Other service activities	0
AUS	Australia	T	Activities of households as employers; undifferentiated goods- and services-producing a	0
AUT	Austria	A01_02	Agriculture, hunting, forestry	0.0014
AUT	Austria	A03	Fishing and aquaculture	0

Example...New Zealand's IT and other information services sector used US(2023)\$5.0539 million in Professional, scientific and technical activities from Australia (most likely consultants)

<https://www.oecd.org/en/data/datasets/inter-country-input-output-tables.html>

(A more useable version is available on LEARN)

Economic Assessment

How

1. Identify the scope of your assessment; what is included and what is excluded
2. Identify the Sector and the Expenditure associated with your sub-systems
3. Trace where & how this money will likely be spent
4. Estimate the economic benefits
5. Communicate your findings

Economic Assessment

How (the hard way)



News

Life

Radio

Podcasts

Video

Te Ao Māori

Pacific

IndoNZ

中文

New Zealand

World

Politics

Pacific

Te Ao Māori

Sport

Business

Country

Local Democracy Reporting

POLITICS / INFRASTRUCTURE

Watch: Govt says billions being spent on infrastructure will create thousands of jobs

12:56 pm on 20 July 2025

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Economic Assessment

The easy way for Steps 2 → 4, only NZ

[Matatihi Impact Calculator](#)

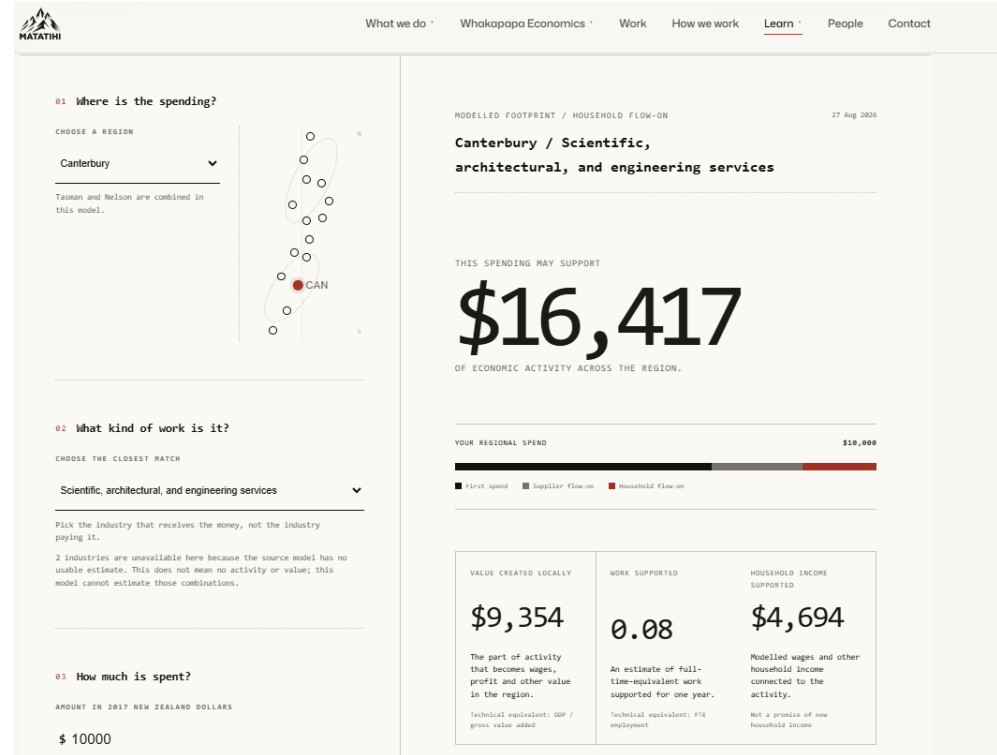
- Identify impacts of economic activities in an NZ region
- Reports only some ANZSIC codes (Industry classifications)
- Uses 2017 NZ\$; need to account for inflation
- **Value Added:** Contribution to (regional) GDP
- **Direct** effects are those occurring within the sector creating the goods or services.
- **Indirect** effects occur in the supply chain, including sectors beyond the one of interest
- **Induced** effects result from households spending a proportion of the additional income

Economic Assessment

The easy way for Steps 2 → 4, only NZ

- \$10,000 of spending in Canterbury on 'Scientific, architectural, and engineering services'
- "Spending multiplier" is 1.6417
→ effect of circular flow of money in an economy

[YouTube video on the spending multiplier](#)
(link also in Learn)



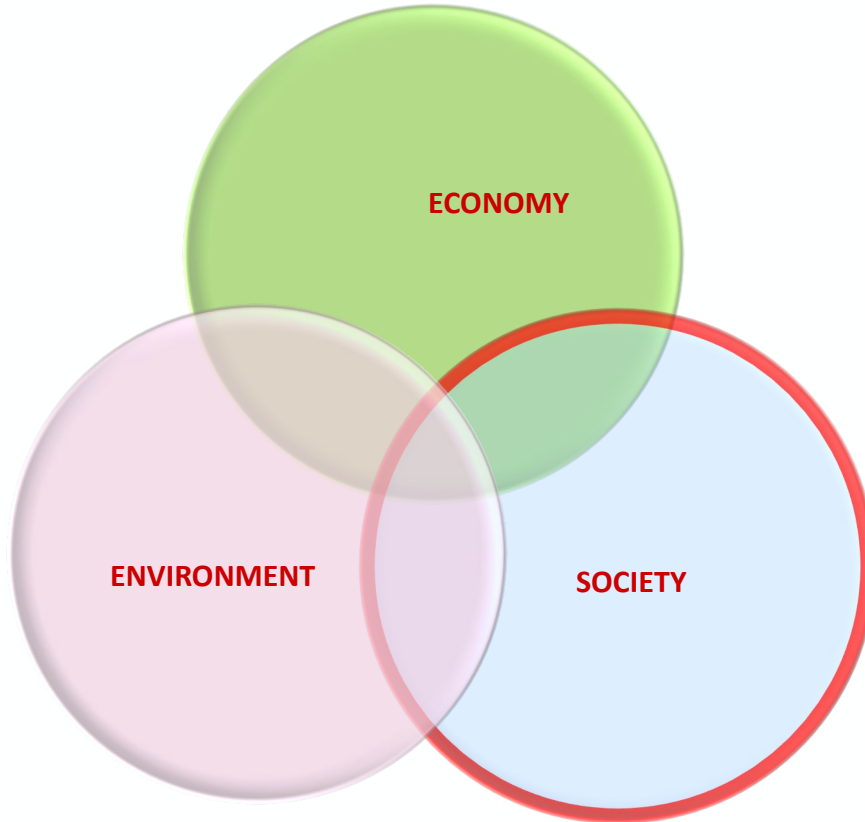
Economic Assessment

For your assignment

- Use the Matatihi calculator to estimate the local and regional economic impacts associated with the operation and use
- Use the Inter-Country Input Output tables to identify and quantify the three largest suppliers (b \$ value) for:
 - Production stage: Assume that all IT equipment is produced in China. Assume all other infrastructure is produced in New Zealand
 - Use stage: Assume electricity generation and supply occurs in New Zealand.
 - End-of-life stage: Identify and cite the assumed locations and processes for the management of waste materials, including for construction materials and IT equipment.
- Induced economic benefits and impacts (qualitative description and justification)
 - indirect impacts from data centre use, such as productivity changes, changes in jobs, etc...
 - *You do not need to quantify this, but it needs to be based on evidence*

What is Social Assessment

Why is this important?



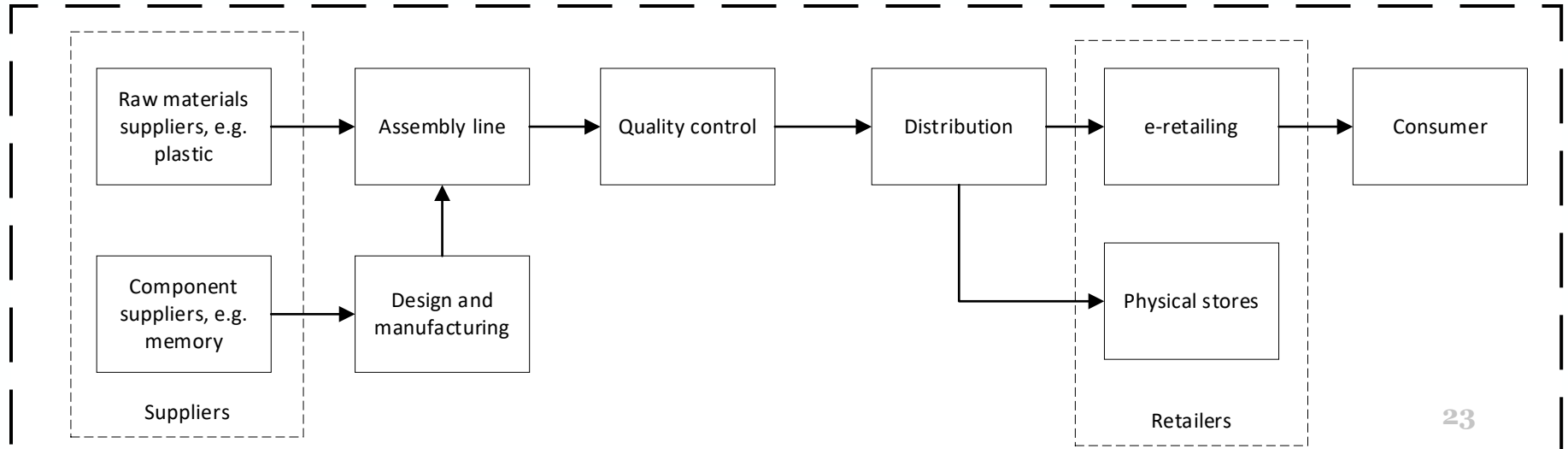
- An evaluation of social and socio-economic outcomes of a product/service
- Doing an economic evaluation first will help with your thinking
- A form of risk assessment; anticipate and manage positive and negative outcomes
- ISO26000 – Social Responsibility

Note: there is no consistently applied approach to assessing social impacts, but we'll use an approach that is similar to risk management (more about this later in ENEL301)

Social assessment

How...

1. Identify the scope of your assessment; what is included and what is excluded
 - A system boundary diagram helps
 - You might need to expand the system, depending on what issues you identify



Social assessment

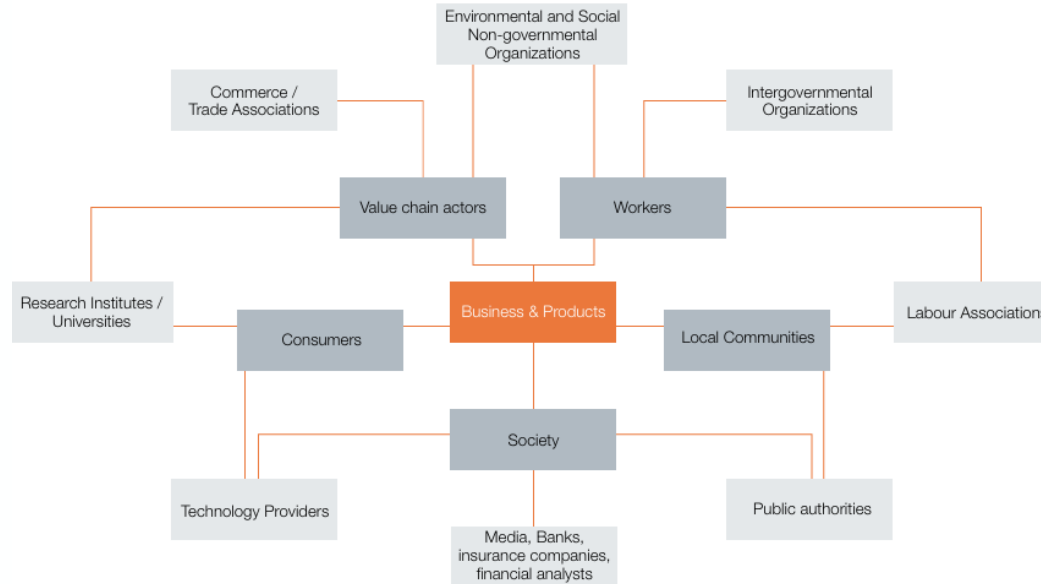
How...

2. Identify and map the key stakeholders (individuals and organisations) who are affected by your system
 - Indirect
 - Supply chain workers (upstream)
 - End of life workers (downstream)
 - Broader stakeholders associated with the whole system (society: local, regional, global)
 - Direct: People directly working on the system, or users (consumers) of the system

Social assessment

How...

2. Identify and map the key stakeholders (individuals and organisations) who are affected by your system



[United Nations. Guidelines for social life cycle assessment of products](#)

Social assessment

How...

3. Identify potential social positive outcomes and negative impacts for the different stakeholders
 - Talk with stakeholders, search the literature, talk with experts
 - Consider what is your company's "social licence to operate" → what would lose people's trust that you were doing the right thing?
 - Systems thinking techniques, e.g. causal loop diagrams, can help
 - Go back to Step 1 if needed

Social assessment

How...

3. Identify potential social positive outcomes and negative impacts

Stakeholder Category	Social impact
Worker	Child labour Salary Working hours Health and safety
Consumer	Consumer privacy Health and safety Consumer law
Local community	Cultural heritage Local employment Health (e.g. arising from pollution)
Society	Economic development Conflict (e.g. war)
Supply chain actors (inc. workers, companies)	Intellectual property

[United Nations. Guidelines for social life cycle assessment of products](#)

Social assessment

How...

3. Identify potential social positive outcomes and negative impacts



social impacts of data centres



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 Create alert

[The Cloud next door: investigating the environmental and socioeconomic strain of datacenters on local communities](#)
WM Ngata, N Bashir, M Westerlaken, L Liote... - Proceedings of the ..., 2025 - dl.acm.org
... 2 – **Social impacts** of datacenters are often framed only in terms of economic benefits, with claims about job creation and community development. However, datacenters do not create ...
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[\[HTML\] The “whole systems” energy sustainability of digitalization: Humanizing the community risks and benefits of Nordic datacenter development](#)
BK Sovacool, P Upham, CG Monyei - Energy Research & Social Science, 2022 - Elsevier
... **impacts** to datacenter development occur at the “whole systems” level? We examine not only **impacts** ... manufacturing of equipment, the laying of **data** cables, the construction of buildings...
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[\[PDF\] Data centres as impermanent infrastructures](#)
J Velkova - Culture machine, 2019 - helda.helsinki.fi
... **data** center, we can come to understand the full extent of their economic and **social** fragility.
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Social assessment

How...

3. Identify potential social positive outcomes and negative impacts

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What are conflict minerals?

"Conflict minerals," as defined by the US legislation, currently include the metals tantalum, tin, tungsten and gold, which are the extracts of the minerals cassiterite, columbite-tantalite and wolframite, respectively. Downstream companies often refer to the extracts of these minerals as 3TG.

- Note: Conflict minerals can be extracted at many different locations around the world including the Democratic Republic of Congo (DRC). The SEC rules define conflict minerals as 3TG metals, wherever extracted. For example, tin extracted in Canada, Russia or Argentina is considered a conflict mineral by definition. In the SEC rule, "DRC conflict-free" is defined as minerals that were extracted and did not directly or indirectly benefit armed groups in the covered countries. Therefore, tin extracted from Canada is considered "DRC conflict-free" under the definitions of the SEC rule.

The internationally recognized OECD Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas, has a broader scope and covers all minerals, not only 3TG.

[Responsible Minerals Initiative](#)

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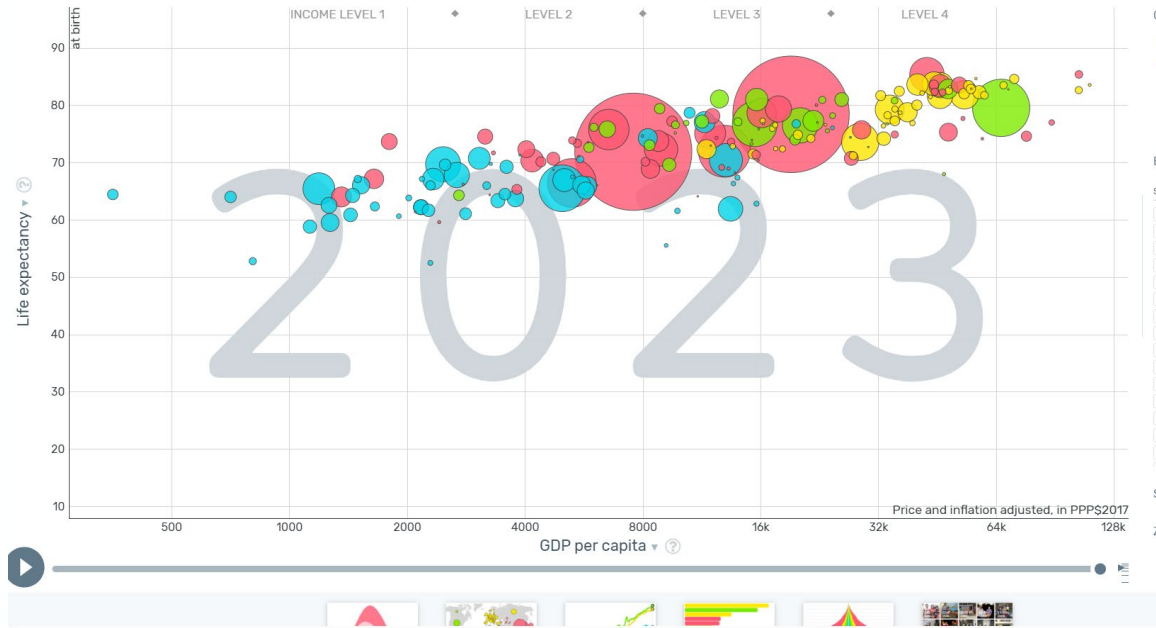
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Social assessment

How...



<https://www.gapminder.org/tools/>

Social assessment

How...

3. Identify potential social positive outcomes and negative impacts



Indicator List:

<https://unstats.un.org/sdgs/indicators/indicators-list/>

- 4.6.1 Youth/adult literacy rate
- 5.b.1 Proportion of individuals who own a mobile telephone, by sex
- 7.1.1 Proportion of population with access to electricity
- 12.5.1 National recycling rate, tons of material recycled

Assignment: You will link social outcomes with the UN SD Goals, Targets, Indicators

Social assessment

How...

4. Analyse benefits and risks (likelihood x consequence) . *Not needed for your assignment. Social LCA is a way of doing this*
5. Identify possible mitigation techniques. *Not needed for your assignment*
6. Communicate findings

But, what do you do if there is a trade-off between social outcomes?

You get increased access to the internet, at the expense of a war over the need for tin

You can actually assign a \$ value on these outcomes → “pricing externalities”

Take-aways

The economy interlinks a lot, including our financial, social and environmental outcomes

Economic Assessment

1. Identify the scope of your assessment
2. Identify the Sector and the Expenditure
3. Trace where & how this money will likely be spent
4. Anticipate the economic change
5. Communicate your findings

* Not needed for your assignment

Social Assessment

1. Identify the scope of your assessment
2. Identify and map the key stakeholders (individuals and organisations) who are affected by your system
3. Identify potential social positive outcomes and negative impacts for the different stakeholders
4. Analyse benefits / risk *
5. Identify possible mitigation techniques *
6. Communicate your findings

Reminders and where you should be at...

Sustainability Assignment. By the end of this week, you should have:

- Met your team, worked out your ways of working, worked out a plan
- Fully comprehended the brief
- Completed some modelling and writing; causal loop diagram
- Have started other modelling: life cycle assessment, economic impacts
- First round of peer-assessment next week
- **Complete the survey on team behaviours (under Assessment → Sustainability Assignment)**
- Next week's lectures: Time Value of Money & Cash Flow
- Next week's workshop: Time Value of Money & Cash Flow. Cost Accounting

Resources

- [NZ System of National Accounts - Input Output Tables \(2024\)](#)
- [Matatihi Impact Calculator](#)
- [Quantifying fiscal multipliers in New Zealand: The evidence from SVAR model](#)
- [United Nations. Guidelines for social life cycle assessment of products](#)
- [OECD Inter-Country Input Output Tables](#)